FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Sickle Cell Disease in Pregnancy

Lecture | Week 3 · Medical disorders & PPH | Obstetrics | Dr Mutamba

Pregnancy (a hypercoagulable, high-oxygen-demand state with extra anaemia and infection risk) makes sickle cell crises more frequent and dangerous and damages the placenta, so women with SCD need pre-conception planning, folic acid 5 mg, malaria prevention, penicillin and vaccination, aspirin, thromboprophylaxis, serial growth scans, and delivery at 38–40 weeks in a warm, well-hydrated, well-oxygenated labour.

Must know before the ward round

1. SCD (HbSS, HbSC, HbS/β-thalassaemia) raises maternal death, pre-eclampsia, VTE, infection, preterm birth, growth restriction and stillbirth several-fold; crises are most common in the third trimester and the puerperium.
2. A sickling (solubility) test cannot tell AS from SS; the genotype needs Hb electrophoresis or HPLC, for her and for her partner (SS × AS gives a 50% chance of an affected baby).
3. Give folic acid 5 mg daily before conception and throughout pregnancy; do NOT give routine iron (only if ferritin shows deficiency). In Uganda remember malaria chemoprevention and an ITN.
4. Folic acid 5 mg antagonises IPTp-SP, so units use daily proguanil 200 mg, or monthly SP with folic acid withheld for 1–2 weeks after each dose. Follow the unit protocol, but never leave her without malaria prevention.
5. Stop hydroxyurea (hydroxycarbamide) at least 3 months before conception (and ACE inhibitors and iron chelators); continue penicillin V prophylaxis and update vaccines.
6. Aspirin 75 mg daily from 12 weeks; LMWH during any antenatal admission and postnatally (7 days after vaginal birth, 6 weeks after caesarean section, RCOG).
7. Painful crisis gets analgesia within 30 minutes (paracetamol, then morphine; NSAIDs only between 12 and 28 weeks; avoid pethidine), fluids, oxygen if SpO_2 is below 95%, and a search for infection and obstetric causes.
8. The crisis workup is FBC with reticulocytes compared with her baseline Hb, mRDT, urinalysis and culture, blood cultures if febrile, group and cross-match, SpO_2 and RR 4-hourly, and a CXR for any chest sign.
9. Falling Hb with very low reticulocytes means an aplastic crisis (parvovirus B19, check the fetus for anaemia); high reticulocytes mean haemolysis, malaria or sequestration.
10. Acute chest syndrome (fever or chest symptoms plus a new CXR infiltrate) is the leading killer; give oxygen, antibiotics (cephalosporin + macrolide), analgesia, incentive spirometry and early transfusion/exchange.
11. Routine prophylactic transfusion is not recommended; transfuse for acute anaemia (Hb below 6 g/dL or a fall of 2 g/dL or more from baseline), ACS, stroke or multi-organ failure.
12. Serial growth scans every 4 weeks from 24 weeks; deliver at 38+0 to 40+0 weeks; in labour keep her warm, hydrated and oxygenated, with continuous CTG and regional analgesia.

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1. Definition

Sickle cell disease (SCD) is an inherited haemoglobin disorder in which **haemoglobin S polymerises when deoxygenated**, turning red cells into rigid "sickles" that **block small vessels (vaso-occlusion)** and **break down early (haemolysis)**.

Pregnancy stacks four extra stresses on top of SCD: **more anaemia, a hypercoagulable state, higher oxygen demand, and more infection**, and the **placenta** itself is a low-oxygen vascular bed where sickling happens easily. The result is more **painful crises, acute chest syndrome, thrombosis, pre-eclampsia, growth restriction, preterm birth and stillbirth**, and a maternal mortality several times higher than that of women without SCD.

Term	Meaning
Sickle cell disease	Umbrella term for genotypes causing clinical sickling: **HbSS** (sickle cell anaemia, most common and most severe in Uganda), **HbSC**, **HbS/β⁰-thalassaemia** (severe), **HbS/β⁺-thalassaemia** (milder)
Sickle cell trait (HbAS)	Carrier state. **Not a disease**: normal Hb, no crises. Mild increase in **UTI** and haematuria; important for **partner counselling**
Vaso-occlusive (painful) crisis	Acute pain from microvascular occlusion and ischaemia in bone marrow, chest, abdomen
Acute chest syndrome (ACS)	New pulmonary infiltrate on CXR plus fever and/or respiratory symptoms (cough, chest pain, tachypnoea, hypoxia)
Sequestration crisis	Sudden pooling of blood in the spleen (or liver) with acute fall in Hb and enlarging organ
Aplastic crisis	Temporary marrow shutdown (usually **parvovirus B19**) with falling Hb and **very low reticulocytes**
Hyperhaemolytic crisis	Accelerated haemolysis: falling Hb, **high reticulocytes**, rising bilirubin and LDH
Baseline (steady-state) Hb	Her usual Hb when well (often 6–9 g/dL in HbSS; higher in HbSC). Always compare to this

> **CLINICAL PEARL**
> The one thing never to forget: **cold, dehydration, hypoxia, infection and acidosis make red cells sickle.** Almost every intervention in SCD pregnancy (warmth, fluids, oxygen, antibiotics, malaria prevention, early analgesia) is about removing one of these triggers.

2. Epidemiology

- About **1 in 7 Ugandans carries the sickle gene** and an estimated **15,000–20,000 babies are born with SCD each year** Verify *births-per-year estimate and source*. With better childhood care (penicillin, hydroxyurea, malaria prevention), more girls with SCD now reach adulthood and pregnancy. You **will** meet them on the ward.

- The **Uganda Sickle Surveillance Study** (2014–2015, dried blood spots from infants nationwide) found **sickle cell trait in about 13%** and **SCD in about 0.7% of infants**, with the highest rates in the **east-central (Busoga), mid-northern and north-eastern** regions and lower rates in the south-west.
- High trait prevalence is the footprint of **malaria**: HbAS protects against severe falciparum malaria, so the gene was selected for.
- Without care, many children with SCD in Africa die before age 5. Uganda now has **SCD clinics** at Mulago (the national sickle cell clinic), regional referral hospitals and some general hospitals, with **hydroxyurea** increasingly available. This means more women conceive **while taking hydroxyurea**.
- Real-world obstacles: **blood shortages** (and alloantibodies making cross-match slow), no HbS-negative/phenotyped blood, limited access to LMWH, no exchange transfusion outside a few centres, limited echo, and women who present late or book at HC IIIs without specialist input.
- Pregnancy outcomes in African cohorts are poorer than in high-income countries: higher rates of maternal death (mainly from ACS, sepsis, severe anaemia and thromboembolism), stillbirth and low birth weight.
- Crises are most common in the **third trimester and the puerperium**.

> **UGANDA CONTEXT**
> Standard Ugandan ANC gives **ferrous sulphate + folic acid (FeFo)** and **IPTp-SP** to everyone. For a woman with SCD, **modify both**: give **folic acid 5 mg daily** (not the low-dose combination), **withhold routine iron** unless iron deficiency is proven, and choose malaria chemoprevention carefully (section 9), because **folic acid at 5 mg or more reduces the effectiveness of SP**.

3. Aetiology and Risk Factors

Cause: autosomal recessive inheritance. A point mutation in the **β-globin gene (codon 6, glutamic acid → valine)** produces HbS. Disease needs two abnormal β-globin alleles: SS, SC (HbC on the other allele), or S with β-thalassaemia.

Inheritance (what to tell the couple)

Parents	Each pregnancy
SS × AA	**100% AS** (trait)
SS × AS	**50% SS**, 50% AS
AS × AS	**25% SS**, 50% AS, 25% AA

Triggers of sickling and crises ("Cold, Dry, Low O_2, Bugs, Acid")

- **Cold** (cold weather, cold IV fluids, cold wards and theatres).
- **Dehydration** (vomiting, hyperemesis, poor intake, prolonged labour).
- **Hypoxia** (pneumonia, atelectasis, opioid hypoventilation, general anaesthesia).
- **Infection**, especially **malaria**, UTI/pyelonephritis, pneumonia, puerperal sepsis.
- **Acidosis**; also stress and exertion (labour).

Risk factors for a severe course in pregnancy

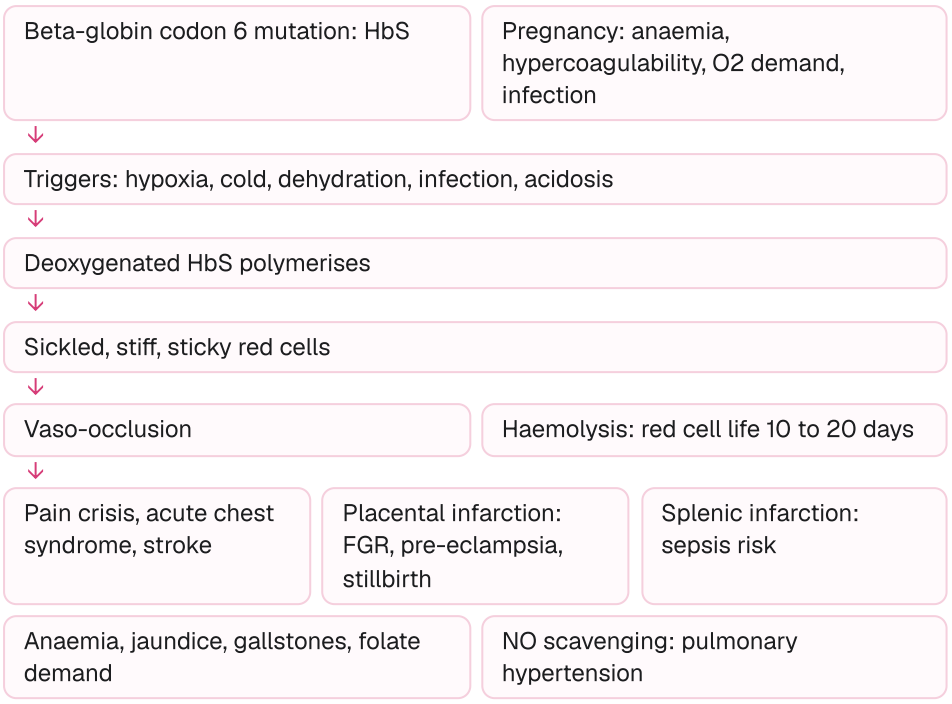
- Previous **acute chest syndrome, stroke, ICU admission or sequestration**.
- Frequent crises in the last year; **worsening crises in this pregnancy**; **twin pregnancy**.
- End-organ damage: **pulmonary hypertension**, cardiomyopathy, nephropathy/proteinuria.
- **Alloantibodies** from previous transfusions (blood hard to find).
- Late booking at a facility without specialist input.
- **HbSC** is not "mild" in pregnancy (sequestration, ACS, VTE, retinopathy).

4. Pathophysiology

The basic mechanism

The β-globin mutation produces HbS. **Deoxygenated HbS polymerises** into long rigid fibres (triggers: hypoxia, acidosis, dehydration, cold, infection, stress). Red cells become **sickled, stiff and sticky**, which causes two processes:

- **Vaso-occlusion**: bone/marrow infarction (pain), acute chest syndrome, stroke, priapism, avascular necrosis of the hip, splenic infarction (**auto-splenectomy → infection with encapsulated bacteria**).
- **Haemolysis** (red cell life 10–20 days, not 120): chronic anaemia, jaundice, pigment gallstones; free Hb scavenges nitric oxide → vasoconstriction, endothelial damage, pulmonary hypertension; high folate demand (marrow working hard).



Why pregnancy makes SCD worse

Physiological change	Consequence in SCD
Plasma volume rises more than red cell mass (dilutional anaemia) plus increased folate demand	Hb falls from an already low baseline; **folate deficiency** causes megaloblastic crisis
Hypercoagulability (rise in fibrinogen and factors VII, VIII, X; fall in protein S; venous stasis)	Adds to the prothrombotic state of SCD, so **VTE risk rises markedly**
Increased oxygen consumption and the **placental bed** (low-oxygen intervillous space)	Sickling in the placenta: **placental infarcts, fibrosis, abruption**
Relative immunosuppression, urinary stasis	More **UTI, pyelonephritis, pneumonia, malaria**, puerperal sepsis, all of which trigger crises
Nausea and vomiting, reduced intake	Dehydration triggers crises
Labour: pain, exertion, dehydration, acidosis, hypothermia, blood loss	**Crises in labour and the puerperium**
Diaphragmatic splinting late in pregnancy and after CS; pain limiting breathing	Basal atelectasis, which predisposes to **acute chest syndrome**

Why the fetus suffers

Sickling in the **uteroplacental circulation** reduces placental blood flow, causing **placental insufficiency**. The fetus gets less oxygen and nutrients, leading to **growth restriction, oligohydramnios, fetal distress and stillbirth**. Maternal severe anaemia and hypoxia during crises add acute insults. Placental vascular disease also underlies the increased **pre-eclampsia** risk. The fetus does **not** sickle in utero, because fetal haemoglobin (HbF) predominates; SCD in the baby shows only after about 6 months as HbF falls.

Feature links

- Bone marrow infarction → ischaemic pain → vaso-occlusive crisis (back, limbs, chest, abdomen).
- Pulmonary vaso-occlusion, atelectasis from splinting or opioids, infection → new infiltrate and hypoxia → acute chest syndrome (fever, cough, chest pain, tachypnoea, falling SpO_2).
- Chronic haemolysis → anaemia → pallor; bilirubin → jaundice and pigment gallstones (RUQ pain).
- Haemolysis → high marrow turnover → folate demand → megaloblastic anaemia if folate is short.
- Splenic infarction → functional asplenia → fever is an emergency (pneumococcus, *Salmonella*).
- Splenic pooling (HbSC, S/β-thal) → enlarging spleen with falling Hb → sequestration and shock.
- Parvovirus B19 → marrow shutdown → falling Hb with very low reticulocytes (aplastic crisis); fetus also infected → fetal anaemia and hydrops.
- Cerebral vaso-occlusion → weakness, speech disturbance, seizures → stroke.
- Placental sickling and infarction → placental insufficiency → small SFH, oligohydramnios, FGR, stillbirth; placental vascular disease → pre-eclampsia.
- Pregnancy hypercoagulability plus SCD → DVT and PE (leg swelling, pleuritic pain; an ACS mimic).
- Free Hb scavenging nitric oxide → pulmonary hypertension → breathlessness, loud P2, TR jet above 2.5 m/s.
- Renal medullary sickling → poor urine concentration (specific gravity unreliable), haematuria (papillary necrosis), proteinuria (nephropathy).
- Recycled haem iron and transfusions → iron overload → no routine iron.

The "why" behind the management

Intervention	Mechanism it targets
Folic acid 5 mg	Chronic haemolysis consumes folate for new red cells
No routine iron	Iron is recycled from haemolysed cells, and transfusions add iron: most women are **iron-overloaded**, not deficient
Penicillin + vaccines	**Functional asplenia** leaves her vulnerable to pneumococcus, meningococcus and *H. influenzae*
Malaria prevention	Malaria causes haemolysis, fever, dehydration and hypoxia: a major crisis trigger and killer in Africa
Aspirin from 12 weeks	Reduces placental vascular disease and pre-eclampsia
LMWH	Counteracts the combined hypercoagulability
Warmth, fluids, oxygen	Prevent deoxygenation, cold-induced vasoconstriction and increased blood viscosity
Incentive spirometry	Prevents atelectasis, the precursor of acute chest syndrome
Stop hydroxyurea before conception	Hydroxyurea is teratogenic in animal studies (it inhibits DNA synthesis)

5. Classification / Staging

By genotype

Genotype	Typical severity	Pregnancy points
HbSS (sickle cell anaemia)	Most common and most severe in Uganda; baseline Hb often 6–9 g/dL; spleen usually auto-infarcted in adults	Crises, ACS, anaemia
HbSC	Higher baseline Hb, few previous crises, so often overlooked	**Not "mild" in pregnancy**: sequestration (spleen often still present), ACS, VTE, **proliferative retinopathy**
HbS/β^0-thalassaemia	Severe (like SS)	As SS; sequestration risk
HbS/β^+-thalassaemia	Milder	Sequestration risk
HbAS (trait)	Not disease	Normal pregnancy care plus urine screening and partner counselling; **not** the SCD protocol

By type of crisis

Crisis	Hb	Reticulocytes	Key feature
Vaso-occlusive (painful)	Near baseline	Baseline	Pain typical of her previous crises
Acute chest syndrome	Often falling	Variable	New CXR infiltrate + fever or respiratory symptoms
Sequestration	Falls fast	High	Enlarging spleen or liver, shock
Aplastic	Falls	**Very low**	Parvovirus B19
Hyperhaemolytic	Falls	High (may fall in severe forms)	Jaundice, high LDH/bilirubin, often after transfusion

6. Clinical Features

The commonest scenario on the ward is a pregnant woman with SCD who arrives **in pain**. In clinic, the woman is usually well, and the job is to set her baseline and plan.

History

Each item has a reason.

Question	Why it matters
Which genotype (SS, SC, S/β-thal)? How was it diagnosed (electrophoresis/HPLC, not a sickling test)?	Severity and specific risks (SC: sequestration, retinopathy, often under-recognised)
Baseline Hb (from her SCD clinic book) and usual symptoms	You need a reference point to recognise acute anaemia. Every Hb today is interpreted against it
This pain: site, onset, severity; **is it like her usual crises?**	Atypical pain = look for another cause (abruption, labour, pyelonephritis, cholecystitis, appendicitis, HELLP, ectopic in early pregnancy)
Analgesia already taken (at home or at the HC): type, dose, time	Avoids overdose and guides the next dose
Trigger: fever, rigors (malaria), cold exposure, dehydration or vomiting, recent travel, dysuria, cough	Find and treat the trigger
Chest symptoms: cough, chest pain, breathlessness	ACS often starts 1–3 days after a painful crisis

Question	Why it matters
Neurological symptoms: weakness, speech disturbance, headache, visual change	Stroke or eclampsia
Obstetric symptoms: gestational age, contractions, bleeding, liquor, **fetal movements**	Labour, abruption, fetal wellbeing
Crises in the last year: number, admissions, ICU, **previous ACS, stroke, sequestration**	Predicts crises in pregnancy and a severe course; previous ACS/stroke may indicate transfusion programmes
Transfusion history and any problems (reactions, "difficult cross-match")	**Alloantibodies** make blood hard to find, delay cross-match and increase haemolytic disease of the newborn risk
Medicines: hydroxyurea (still taking it?), penicillin, folic acid 5 mg, aspirin, malaria chemoprevention, LMWH, iron chelators, ACE inhibitors, analgesics (including opioid use), iron (should not be routine)	Hydroxyurea/chelators/ACEi must stop; penicillin and folic acid must continue
Allergies (penicillin, opioids)	Drug choice
Vaccination history (pneumococcal, meningococcal, HiB, hepatitis B, influenza/COVID)	Asplenic protection
End-organ damage: breathlessness/exercise tolerance (pulmonary hypertension, cardiomyopathy), kidney disease/proteinuria, eye problems, leg ulcers, avascular necrosis of hip, gallstones	Baseline complications change pregnancy risk and delivery planning
Partner's Hb status	If he is AS, there is a 50% chance of an affected baby (AS × SS); if AA, the baby will be AS
Previous pregnancies: miscarriage, IUGR, pre-eclampsia, stillbirth, CS, postpartum crises	Recurrence risks
Malaria prevention: ITN use, chemoprophylaxis	Crisis trigger

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), with the obstetric abdominal examination. A woman in severe pain gets analgesia and the ABCDE assessment first (section 9); complete the examination once the first dose is given.

Introduction, consent and preparation

- Introduce yourself, confirm her identity and gestation, explain, gain consent; chaperone for abdominal and vaginal examination.
- **Hand hygiene**.
- **Position and exposure**: comfortable, semi-recumbent with **left lateral tilt**; keep her **warm** (blankets; expose only what you examine).

General inspection and vital signs

Sign	Mechanism	What it means
Distress, pain score (0–10)	Vaso-occlusion	Guides analgesia
Pallor	Chronic haemolysis, acute fall	Compare Hb with baseline
Jaundice	Haemolysis; hepatic crisis, cholecystitis, HELLP	LDH, bilirubin, LFTs
Dehydration (dry mucosae)	Vomiting, poor intake	Crisis trigger: fluids

Sign	Mechanism	What it means
Leg ulcers	Chronic haemolysis, vaso-occlusion	Long-standing severe disease
Frontal bossing, maxillary hyperplasia, poor dental hygiene	Marrow expansion; infection source	Chronic disease

Vital signs: temperature (infection), RR and **SpO_2** (ACS), HR, **BP** (pre-eclampsia; SCD patients often have a low baseline BP, so a "normal" 130/85 may be relatively high), pain score.

Hands, face and neck

- Hands: palmar pallor, swollen tender digits (dactylitis is rare in adults).
- Face: conjunctival pallor, scleral **jaundice**, facial or speech asymmetry (stroke).
- Neck: JVP (fluid overload, pulmonary hypertension), neck stiffness (meningitis).

Chest and heart

Crackles, bronchial breathing, dullness at the bases, reduced air entry (ACS, pneumonia); **splinting** from rib or back pain; signs of pulmonary hypertension (loud P2, right heart failure). Repeat the chest examination every 4 hours during a crisis.

Obstetric abdominal examination

- **Inspection**: size for dates, scars.
- **Palpation**: **spleen** (usually impalpable in adults with HbSS because of auto-infarction; **a palpable, enlarging spleen with falling Hb means sequestration**, more likely in HbSC or S/β-thal), liver (hepatic sequestration), RUQ tenderness (cholecystitis, hepatic crisis), renal angle tenderness (pyelonephritis), **uterine tone and tenderness** (abruption), contractions (preterm labour).
- **Fundal height**: small for dates → IUGR.
- **Lie and presentation**.
- **Engagement** (fifths palpable).
- **Auscultation of the fetal heart** (CTG after 28 weeks).

Neurological

If any headache, weakness or speech change: focal deficit, neck stiffness, GCS (stroke, eclampsia, meningitis).

Completing the examination

- **Bones and joints**: tender, swollen areas in a painful crisis; does it match her usual pattern? Hips (avascular necrosis).
- **Legs**: calf swelling (DVT), leg ulcers. DVT and PE mimic ACS.
- **Bedside tests**: SpO_2, urine dipstick, HemoCue, mRDT (section 8).
- Thank her, cover her, hand hygiene, summarise.

Document key positives and negatives

Area	Model documentation
General	"Pale, mildly jaundiced, dry mucosae, T 38.4 °C"
Vital signs	"RR 20, SpO_2 97% on air, P 108, BP 118/72 (booking 100/60)"
Chest	"Chest clear, good air entry at both bases" (repeat every 4 hours)
Abdomen	"Spleen not palpable, liver 2 cm, no renal angle tenderness"
Obstetric	"SFH 30 cm at 32 weeks, uterus soft, non-tender, FHR 146"

Area	Model documentation
Neurological	"No focal deficit, GCS 15"

Red flags

RED FLAG

- **Chest pain, cough, fever, breathlessness or SpO_2 below 95%**: acute chest syndrome until proven otherwise (CXR, oxygen, antibiotics, senior review, early transfusion).
- **Sudden drop in Hb** (2 g/dL or more below baseline, or below 6 g/dL): sequestration, aplastic crisis (check reticulocytes), hyperhaemolysis or bleeding (abruption, PPH).
- **Enlarging spleen or liver with falling Hb**: sequestration: urgent transfusion.
- **Weakness, facial droop, speech disturbance, severe headache, seizures**: stroke (exchange transfusion) or eclampsia.
- **Fever 38 °C or more**: overwhelming sepsis risk (asplenia) and malaria: cultures, mRDT, IV antibiotics now.
- **Pain not controlled by opioids, or atypical pain**: think abruption, labour, ectopic, pyelonephritis, appendicitis, cholecystitis, pancreatitis, VTE, or HELLP.
- **Leg swelling or pleuritic pain**: DVT/PE (an ACS mimic).
- **Reduced fetal movements** or falling SFH: IUGR/fetal compromise.

Thinking like a clinician

- **"It's just another crisis" is the most dangerous phrase on the ward.** A woman with SCD in pregnancy who has abdominal pain may have **abruption, labour, pyelonephritis, ectopic pregnancy, appendicitis, cholecystitis or HELLP**. Examine the abdomen properly, check the fetal heart, urine and BP every time.
- **Every painful crisis in pregnancy is a potential acute chest syndrome**: ACS often develops 1–3 days into an admission for pain, especially after opioids (hypoventilation) and with rib/back pain. Check SpO_2 and RR 4-hourly; encourage incentive spirometry.
- **Compare every Hb to her baseline.** A Hb of 7 g/dL is reassuring in a woman whose baseline is 7, and alarming in a woman with HbSC whose baseline is 11.
- **Reticulocytes help separate acute anaemias**: low reticulocytes = aplastic (parvovirus); high = haemolysis or sequestration.
- **Fever is an emergency** because of asplenia. Do not wait for the malaria result before giving antibiotics if she is septic.
- **HbSC is not "mild" in pregnancy**: these women may have near-normal Hb and few previous crises, so they are overlooked, yet they are at risk of sequestration, ACS, VTE and retinopathy.
- **Sickle trait (AS) is not disease**: treat as normal pregnancy (with urine screening and partner counselling). Do not give her the SCD protocol.
- **A "normal" BP may be high for her**: compare with her booking BP.
- **Common student mistakes**:
  1. Prescribing routine FeFo (iron) to a woman with SCD.
  2. Forgetting that **5 mg folic acid interferes with SP**; failing to plan malaria chemoprevention.
  3. Giving **pethidine** for crisis pain (risk of seizures from its metabolite norpethidine).
  4. Giving NSAIDs in the first trimester or after 28 weeks.
  5. Over-hydrating (pulmonary oedema, especially with pre-eclampsia or cardiac disease); monitor fluid balance.
  6. Forgetting thromboprophylaxis after delivery.

7. Differential Diagnoses

Pain in a pregnant woman with SCD

Differential	Distinguishing clinical feature	Distinguishing investigation
Vaso-occlusive crisis	Typical of her previous crises (limbs, back, chest, abdomen), triggers (cold, infection, dehydration), normal uterine tone	Diagnosis of exclusion: FBC, retics, infection screen
Placental abruption	Constant abdominal pain, **woody-hard tender uterus**, vaginal bleeding (may be concealed), fetal distress/absent FHR, shock out of proportion	Clinical, CTG, ultrasound (limited sensitivity), Hb, clotting
Preterm or term labour	Regular painful contractions, cervical change	Speculum/VE, CTG
Pyelonephritis	Fever, rigors, loin pain, dysuria	Urinalysis, urine culture
Acute chest syndrome	Chest pain, cough, fever, tachypnoea, hypoxia, new infiltrate	CXR, SpO_2, blood gas
Pulmonary embolism	Sudden pleuritic pain, breathlessness, tachycardia, clear chest on auscultation, leg swelling	CTPA/V/Q where available, Doppler legs, ECG
Pre-eclampsia/HELLP	Epigastric/RUQ pain, rising BP (relative to her low baseline), proteinuria, low platelets, raised liver enzymes	BP, urine protein, FBC, LFTs
Cholecystitis / biliary colic (pigment gallstones)	RUQ pain after meals, Murphy's sign, fever	Ultrasound, LFTs
Hepatic sequestration / hepatic crisis	Painful enlarging liver, rising bilirubin, falling Hb	LFTs, FBC, ultrasound
Appendicitis	Migratory pain, right-sided (higher in pregnancy), fever	Clinical, ultrasound
Ectopic pregnancy (early pregnancy)	Amenorrhoea, unilateral pain, bleeding, shock	Pregnancy test, ultrasound
Malaria	Fever, rigors, headache, haemolysis	mRDT, blood smear

Acute anaemia in SCD

Cause	Reticulocytes	Other features
Aplastic crisis (parvovirus B19)	**Very low**	Recent viral illness; fetal risk of anaemia/hydrops
Splenic/hepatic sequestration	High	Enlarging tender organ, shock
Hyperhaemolysis (often post-transfusion)	High (may fall in severe forms)	Jaundice, high LDH/bilirubin, Hb below pre-transfusion level
Malaria haemolysis	High	Positive mRDT/smear
Folate deficiency (megaloblastic)	Low	Macrocytes, hypersegmented neutrophils; poor folate adherence
Bleeding (abruption, PPH, GI)	Rise after days	Visible or concealed blood loss

8. Clinical Workup

The crisis workup has three aims, in this order: **control the pain fast**, **find what is dangerous** (acute chest syndrome, severe anaemia, sepsis, stroke, an obstetric emergency), and **find the trigger** (malaria, infection, dehydration, cold). Availability reflects the source notes Verify *compiled from the source notes, not an MoH list.*

Bedside

Test	Why / expected finding	Available at
Pulse oximetry (on arrival, then at least 4-hourly; continuous if falling)	A falling SpO_2 is the earliest sign of ACS	HC IV upwards; often missing at HC III
Pain score (every 30 minutes until controlled)	Guides analgesia titration	All levels
mRDT (± blood smear)	Malaria is the commonest trigger in Uganda. A positive HRP2 mRDT may stay positive for weeks after treatment	All levels
HemoCue Hb	A quick comparison with baseline while the FBC is pending	HC III upwards
Urine dipstick	Nitrites/leucocytes (UTI, pyelonephritis), protein (pre-eclampsia, sickle nephropathy), blood (papillary necrosis). Specific gravity is unreliable for hydration because SCD impairs urine concentration	All levels
RBS	Hypoglycaemia in a sick, vomiting woman	HC III upwards
Fetal heart / CTG (CTG after 28 weeks)	Fetal compromise from maternal hypoxia, anaemia or abruption	All levels (CTG at hospitals)
Point-of-care ultrasound	Fetal heart, liquor, placenta; spleen size if you are trained	HC IV and hospitals
Incentive spirometer	Start it now (10 breaths every 2 hours while awake); poor effort = pain limiting breathing	Hospitals (a balloon or bottle-blowing can substitute Verify *local practice*)

EXAM FAVOURITE

A **sickling test (sodium metabisulphite) or solubility test is positive in both HbAS and HbSS**, so it cannot diagnose sickle cell disease. The genotype needs **Hb electrophoresis or HPLC**. A woman with trait (AS) should not be put on the SCD protocol.

Basic lab

Test	When	Expected / abnormal finding	How it changes management (and where)
FBC	Booking, each trimester, every crisis (daily if severe)	Hb vs baseline; **fall of 2 g/dL or more, or Hb below 6 g/dL** = acute anaemia; raised WBC is common in crisis (not always infection)	Transfusion decision. All hospitals, most HC IVs
Reticulocyte count	Every acute anaemia; booking	Usually raised at baseline in HbSS; **very low = aplastic crisis**; high = haemolysis/sequestration/malaria	Separates the causes of acute anaemia. Hospital laboratories
Blood film	Acute anaemia, fever	Sickle cells, target cells, Howell–Jolly bodies, **malaria parasites**, macrocytes (folate deficiency)	Confirms malaria and folate problems
Group and cross-match with antibody screen (extended phenotype where possible)	Every admission, early	Alloantibodies = delay	Warn the blood bank early; request HbS-negative, **ABO, Rh (C, c, E, e) and Kell**-matched blood where available

Test	When	Expected / abnormal finding	How it changes management (and where)
Urine culture	Booking, each trimester, any crisis	Significant bacteriuria	Treat ASB/UTI (a crisis trigger)
U&E / creatinine, LFTs	Booking, crises	Sickle nephropathy; raised ALT/bilirubin (hepatic crisis, cholecystitis, HELLP)	Drug dosing (NSAIDs), fluid plan, pre-eclampsia baseline
Urine protein:creatinine ratio	Booking; new hypertension	Proteinuria from nephropathy (baseline) vs new pre-eclampsia	Interpret later BP/proteinuria against baseline
Serum ferritin	Booking (if iron considered)	Low = true iron deficiency; high = overload (or inflammation)	Iron only if deficient
HIV, syphilis, HBsAg, hepatitis C (if transfused)	Booking	Infections	Routine triple-elimination testing and transfusion-related infection

Imaging

Test	When	Expected finding	How it changes management (and where)
Chest X-ray (with abdominal shielding)	Any chest pain, cough, fever with tachypnoea, **SpO_2 below 95%**	**New infiltrate = acute chest syndrome**; effusion; oedema	ACS protocol: oxygen, ceftriaxone + macrolide, transfusion. The fetal dose is tiny; do not withhold it. Hospitals
Obstetric ultrasound	Dating, anomaly scan (18–22 weeks), **growth every 4 weeks from 24 weeks**; crisis with abdominal pain	IUGR, oligohydramnios, abnormal umbilical artery Doppler; retroplacental clot (ultrasound misses many abruptions)	Timing of delivery; closer fetal surveillance. HC IV and hospitals
Umbilical artery Doppler (when growth falls off)	Growth falling	Raised resistance, absent/reversed end-diastolic flow	Placental insufficiency: guides timing of delivery. RRH/NRH
Abdominal ultrasound	RUQ pain	Gallstones, cholecystitis, hepatomegaly	Surgical review
Leg Doppler ± CTPA/V/Q	Leg swelling, pleuritic pain with a clear CXR	DVT, PE	Treatment-dose LMWH. CTPA only at national referral / private centres

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management (and where)
LDH, bilirubin	Acute anaemia, jaundice	Rising = haemolysis (malaria, hyperhaemolysis, delayed transfusion reaction); LDH also rises in HELLP	Haemolysis vs sequestration vs HELLP. Regional/national referral mostly
Blood cultures	Temperature 38.0 °C or more	Pneumococcus, *Salmonella*, gram-negatives	Antibiotic choice; asplenia makes sepsis rapid. Regional/national referral
Blood gas	SpO_2 below 94% or distress	Hypoxaemia, acidosis	HDU/ICU, exchange transfusion. Few referral units

Test	When	Expected / abnormal finding	How it changes management (and where)
Parvovirus B19 serology	Aplastic crisis	IgM positive	**Fetal anaemia risk**: ultrasound for hydrops, MCA Doppler where available
Hb electrophoresis / HPLC (woman and partner)	Pre-conception or booking	Genotype, HbF %	Confirms SCD; partner counselling. Mulago/CPHL and some regional hospitals
Echocardiography	Booking if breathless or known heart/lung disease	**TR jet velocity above 2.5 m/s** = pulmonary hypertension	Very high maternal risk: joint cardiac/obstetric care. Referral hospitals
Retinal exam	Booking (especially HbSC)	Proliferative retinopathy	Ophthalmology referral

Putting it together

Model one-line summary

> "A 24-year-old G2P1 with HbSS (baseline Hb 8.0 g/dL, previous ACS in 2023) at 32 weeks, admitted with a vaso-occlusive crisis of the back and thighs triggered by malaria, now on day 2 with a new fever, cough, SpO_2 92% on air and right basal crackles."

Problem list

1. **Acute chest syndrome** (suspected, to confirm on CXR) complicating a vaso-occlusive crisis.
2. Malaria (mRDT positive) as the trigger.
3. **Acute-on-chronic anaemia**: Hb 6.4 g/dL (fall of 1.6 g/dL from baseline) with high reticulocytes (haemolysis).
4. Pregnancy at 32 weeks: fetal wellbeing, growth, VTE risk.

Working diagnosis: HbSS in pregnancy with acute chest syndrome following a malaria-triggered painful crisis.

Model case presentation (ward round)

> "Ms A is a 23-year-old primigravida with **HbSS** (baseline Hb 7.5 g/dL, two admissions for painful crises last year, no previous ACS or stroke, never transfused), at **30 weeks**, admitted yesterday with a **vaso-occlusive crisis** affecting her lower back and thighs, triggered by **malaria** (mRDT positive). She received **morphine within 30 minutes**, now on **oral morphine 10 mg 4-hourly** and paracetamol, IV fluids 3 L/day, and **IV artesunate**. Today her pain score is 4/10, temperature 37.4 °C, **SpO_2 97% on air**, RR 18, BP 110/70. Hb is **6.9 g/dL** (baseline 7.5), reticulocytes raised. Fetal heart 142 with a reactive CTG; the last growth scan at 28 weeks was on the 30th centile. She is on **folic acid 5 mg, penicillin V 250 mg twice daily, aspirin 75 mg** and **enoxaparin 40 mg daily**. **Plan**: complete antimalarials, step down analgesia, incentive spirometry, watch for ACS (4-hourly SpO_2/RR), repeat Hb tomorrow, growth scan at 32 weeks, and discuss malaria chemoprevention on discharge."

Worked example 1: the crisis blood panel (the 24-year-old above)

Result (day 1)	Her baseline	Interpretation
Hb **6.4 g/dL**	8.0	Fall of 1.6 g/dL: below the 2 g/dL transfusion trigger, but close; repeat daily
Reticulocytes **high** (above her usual)	Raised	Marrow is responding: **haemolysis**, not aplasia

Result (day 1)	Her baseline	Interpretation
LDH and bilirubin **raised**		Haemolysis (malaria + sickling)
WBC 18 × 10^9/L		Common in crisis; does not by itself prove bacterial infection
Platelets 250 × 10^9/L, ALT normal		Argues against HELLP
mRDT **positive**		**Trigger found**: treat malaria (IV artesunate if severe)
Urine: nitrites negative		UTI unlikely

Day 2: T 38.5 °C, cough, RR 28, **SpO_2 92%** on air, right basal crackles. **CXR: new right lower-lobe infiltrate.** This is **acute chest syndrome**.

Action: call the senior and anaesthetist; **oxygen to SpO_2 94–98%**; **IV ceftriaxone 2 g daily + azithromycin 500 mg daily** (or erythromycin); continue antimalarials; analgesia enough to breathe deeply; **incentive spirometry**; careful fluids with a balance chart; LMWH; **cross-match**: top-up transfusion now that she is hypoxic with a falling Hb, **exchange transfusion** if she worsens or has multilobar disease (Mulago/major centres); CTG. **Delivery does not treat ACS.**

Worked example 2: two acute anaemias

	Woman X (HbSS, 26 weeks)	Woman Y (HbSC, 30 weeks)
Hb (baseline)	**5.2 g/dL** (7.5)	**7.0 g/dL** (11.0)
Reticulocytes	**Very low**	High
Spleen	Not palpable	**Palpable, tender, enlarging**
Other	Flu-like illness last week	Tachycardic, hypotensive
Diagnosis	**Aplastic crisis** (probable parvovirus B19)	**Splenic sequestration**
Action	Transfuse (Hb below 6 and a fall of more than 2); parvovirus serology; **fetal scan for hydrops / MCA Doppler**	**Urgent transfusion** (aim to restore, not exceed, baseline: blood returns from the spleen); senior and haematology input

CLINICAL PEARL

Note woman Y: an Hb of 7.0 g/dL "looks like" an ordinary SCD Hb, but for her (HbSC, baseline 11) it is a **fall of 4 g/dL**. Always write the **baseline next to today's Hb** in your notes.

9. Management

Immediate / emergency

First 5 minutes (woman with SCD arriving in pain)

1. **ABCDE and vital signs**: RR, **SpO_2**, pulse, BP, temperature, GCS, and a **pain score** (0–10). Listen to the **fetal heart**.
2. **Analgesia within 30 minutes of arrival**: paracetamol 1 g plus **morphine 2.5–5 mg IV** (or SC) titrated every 20–30 minutes for severe pain. Do not wait for the doctor's full clerking or the lab results. **No pethidine.**
3. **Oxygen** if SpO_2 is **below 95%** (RCOG uses 94%).

4. **IV access** with blood taken at the same time: **FBC, reticulocytes, group and cross-match (with antibody screen), mRDT/blood smear, blood cultures if febrile**, U&E and LFTs where available.
5. **Warm fluids**: oral if she can drink; otherwise IV 0.9% saline or Ringer's lactate. Keep her warm.
6. **Call a senior now** if any of the following:
   - **SpO_2 below 95%, RR above 24, chest pain, new cough or crackles**: acute chest syndrome.
   - **Hb below 6 g/dL, or 2 g/dL or more below her baseline**, or an enlarging spleen or liver: acute anaemia.
   - **Temperature 38.0 °C or more with tachycardia or hypotension**: sepsis. Give **ceftriaxone 2 g IV** after cultures; do not wait for the mRDT.
   - **Weakness, facial droop, speech change, fits or severe headache**: stroke or eclampsia (check BP and urine protein).
   - **Constant abdominal pain, a hard tender uterus, bleeding or an abnormal fetal heart**: abruption.

A. Vaso-occlusive (painful) crisis: step by step

1. **Assess rapidly** (ABCDE, pain score, SpO_2, temperature, BP, fetal heart). **Exclude obstetric and surgical causes** of pain.
2. **Analgesia within 30 minutes of arrival**, then reassess every 30 minutes until controlled (doses in the drug table): paracetamol for all; **morphine** first-line opioid; NSAIDs **only between 12 and 28 weeks**; codeine or tramadol for moderate pain or step-down; **avoid pethidine** (norpethidine causes seizures).
3. **Fluids**: oral **about 3 L/day** if she can drink; otherwise **IV 0.9% saline or Ringer's lactate** (e.g. 1 L over 4–6 hours then reassess). Keep a **fluid balance chart**; avoid overload.
4. **Oxygen** if **SpO_2 below 95%** (RCOG uses 94%); nasal prongs or mask.
5. **Keep her warm**; avoid cold IV fluids and cold wards.
6. **Look for and treat triggers**: FBC, retics, **mRDT/blood smear**, urinalysis and culture, blood cultures if febrile, CXR if chest symptoms. **Fever: start IV ceftriaxone 2 g once daily** (after cultures) and **treat malaria** with IV artesunate (2.4 mg/kg at 0, 12, 24 hours then daily) if positive and severe, or oral artemether-lumefantrine for uncomplicated malaria (check the UCG advice for the first trimester).
7. **Thromboprophylaxis** (LMWH) and mobilise when possible.
8. **Incentive spirometry** (10 breaths every 2 hours while awake) to prevent ACS.
9. **Monitor** 4-hourly: pain, sedation, RR, SpO_2, temperature, BP, fetal heart. **Daily FBC** in severe crises.
10. **Fetal assessment**: CTG after 28 weeks (a fetal heart abnormality may reflect maternal hypoxia or abruption).
11. Continue folic acid, penicillin, aspirin; step down to oral analgesia as pain settles.

B. Acute chest syndrome (emergency)

Features: fever, cough, chest pain, tachypnoea, **SpO_2 falling**, crackles, **new infiltrate on CXR**. Can follow a painful crisis, infection or delivery. Differential: pneumonia, PE, pulmonary oedema.

1. **Call senior help** (obstetrician, physician/haematologist, anaesthetist); HDU/ICU.
2. **Oxygen** to keep SpO_2 **94–98%**; sit up; CPAP/ventilation if failing.
3. **CXR** (with abdominal shielding), blood gas, FBC, retics, cultures, cross-match.
4. **Antibiotics**: **IV ceftriaxone 2 g once daily + oral azithromycin 500 mg daily or erythromycin 500 mg 6-hourly** (covers atypical organisms).
5. **Analgesia** (enough to allow deep breathing, not so much that she hypoventilates) and **incentive spirometry**.
6. **Transfusion**: **top-up** if Hb has fallen; **exchange transfusion** if severe hypoxia, worsening despite treatment, multilobar disease, or Hb high enough that top-up would cause hyperviscosity.
7. **Careful fluids** (avoid overload), LMWH, consider **PE** as the alternative diagnosis.
8. Fetal monitoring; **delivery does not treat ACS** and should only be considered for obstetric reasons or as part of senior-led resuscitation.

C. Acute anaemia (sequestration, aplastic, hyperhaemolytic)

1. **FBC and reticulocytes**, bilirubin, LDH, parvovirus serology, malaria test, examine spleen and liver.
2. **Transfuse** (top-up) for Hb below 6 g/dL or a fall of 2 g/dL or more, or symptomatic anaemia. In sequestration, transfuse cautiously (blood returns from the spleen as it resolves; aim to restore, not exceed, baseline).
3. **Parvovirus B19** (aplastic crisis): also risk of **fetal anaemia and hydrops**: arrange fetal ultrasound (and MCA Doppler where available).
4. **Hyperhaemolysis** after a transfusion: stop further transfusion unless life-threatening; seek haematology advice (steroids, IVIG).

D. Stroke

Sudden weakness, speech disturbance or seizures: stabilise, exclude eclampsia (BP, proteinuria, $MgSO_4$ if in doubt), urgent imaging, and **exchange transfusion** (or top-up if exchange not available) under specialist care.

Definitive

1. Pre-conception care (SCD clinic + obstetrician)

1. **Counsel about risks** to her and the baby, and plan pregnancy when she is well.
2. **Test the partner** (Hb electrophoresis). If he is AS or has SCD, offer genetic counselling (prenatal diagnosis is available in few centres).
3. **Start folic acid 5 mg orally once daily** (continue throughout pregnancy).
4. **Stop hydroxyurea at least 3 months before conception**. If she becomes pregnant on it, stop it, reassure (the absolute risk appears low) and arrange a detailed anomaly scan; do not advise termination on this basis alone.
5. **Stop ACE inhibitors/ARBs** (used for proteinuria) and **iron chelators** (deferasirox, desferrioxamine).
6. **Continue penicillin V prophylaxis**. Update **vaccines**: pneumococcal, meningococcal, *H. influenzae* type b, hepatitis B, and seasonal influenza/COVID where available.
7. **Screen for end-organ damage**: echo (PHT), renal function and proteinuria, retinal screening, red cell antibodies, iron status.
8. **Optimise**: treat infections, malaria prevention, dental care, blood pressure record.

2. Antenatal care

Book early (first trimester) at a **hospital with obstetric and haematology/physician support**. **Visits every 2–4 weeks**, more frequently in the third trimester. At every visit: BP, urinalysis, SFH, fetal heart, Hb (as needed), crisis/infection screening questions.

Routine antenatal package (doses in the drug table): folic acid 5 mg daily; penicillin V 250 mg twice daily; aspirin 75 mg from 12 to 36 weeks; malaria chemoprevention plus ITN; **iron only if ferritin-proven deficiency** (avoid routine FeFo); LMWH during any antenatal admission (consider antenatal prophylaxis if extra VTE risk factors); tetanus-diphtheria and other routine ANC; mebendazole after the first trimester.

DRUG DOSING

Malaria chemoprevention in SCD pregnancy: the SP–folic acid problem. WHO advises that **folic acid 5 mg daily or more should not be given with IPTp-SP**, because high-dose folate antagonises SP's antifolate action. Units manage this differently:

- **Daily proguanil 200 mg** (safe in pregnancy; give with the folic acid) plus ITN; or
- **Monthly IPTp-SP from 13 weeks**, with folic acid 5 mg **withheld for about 1–2 weeks after each SP dose**; or
- Other regimens per the SCD/haematology team.

Do not simply omit malaria prevention: malaria is a leading trigger of crises and death. If she has HIV and takes co-trimoxazole, **no SP**. **Follow your unit protocol** and document the choice.

Fetal surveillance

- **Dating/viability scan** in the first trimester.
- **Anomaly scan at 18–22 weeks** (detailed if hydroxyurea exposure).
- **Growth scans every 4 weeks from 24 weeks** (biometry, liquor, umbilical artery Doppler if growth falls off).
- **Fetal movement awareness**; CTG when indicated.

Transfusion in pregnancy

- **Routine prophylactic transfusion is not recommended** for all pregnant women with SCD.
- **Indications for top-up transfusion**:
  - **Acute anaemia**: Hb **below 6 g/dL** or a **fall of 2 g/dL or more from baseline** (sequestration, aplastic crisis, hyperhaemolysis);
  - **Symptomatic anaemia** or preparation for **caesarean section** when Hb is low;
  - **Acute chest syndrome** with hypoxia, **stroke** or **multi-organ failure** (these usually need **exchange transfusion** where available);
  - Consider a **regular transfusion programme** for women with **previous severe complications** (e.g. previous ACS, stroke on transfusion), **worsening crises in this pregnancy**, or **twin pregnancy**, on specialist advice.
- **Blood requirements**: **HbS-negative**, matched at least for **ABO, Rh (C, c, E, e) and Kell** where possible, CMV-negative. Check for **alloantibodies** (and warn the lab early). Do not over-transfuse: aim for a **Hb no higher than about 10–11 g/dL** (hyperviscosity triggers crises).

3. Timing and mode of delivery

1. **Deliver in a hospital** with blood bank, theatre, anaesthesia, HDU and neonatal care.
2. **Timing**: with normal fetal growth, offer **induction of labour (or CS if indicated) between 38+0 and 40+0 weeks** (RCOG). Earlier for complications (IUGR, pre-eclampsia, recurrent crises).
3. **SCD is not an indication for caesarean section**; vaginal birth is preferred. CS only for obstetric indications.
4. Inform the **anaesthetist, haematologist/physician, neonatologist and blood bank** on admission; **cross-match** early (antibodies may delay).

4. Intrapartum care: "warm, wet, well-oxygenated, watched"

1. **Warm**: warm labour room, warm IV fluids, blankets; avoid hypothermia in theatre.
2. **Hydrated**: oral fluids; **IV fluids** if not drinking well or vomiting; **fluid balance chart** (risk of overload if pre-eclampsia or cardiac disease).
3. **Oxygenated**: **continuous pulse oximetry**; oxygen if **SpO_2 94% or below**.
4. **Pain relief**: **epidural/regional analgesia is encouraged** (reduces stress, and is ready for CS); avoid pethidine; **avoid general anaesthesia** where possible (hypoxia, ACS risk).

5. **Continuous CTG** (EFM): higher risk of fetal compromise and stillbirth.
6. **Avoid prolonged labour** (dehydration, exhaustion, acidosis). Partograph action line: act early. Consider operative delivery for delay.
7. **Infection**: prompt treatment of any fever (blood cultures, mRDT, antibiotics).
8. **Third stage**: active management; be ready for **PPH** (she has low reserve: a normal blood loss can make her severely anaemic).
9. **Caesarean section**: regional anaesthesia, warmth, oxygen, thromboprophylaxis, early mobilisation, good analgesia for deep breathing; **transfuse beforehand if Hb is low**.

5. Postpartum care

1. **Crisis risk remains high for 1–2 weeks** (about a quarter of women): continue **warmth, hydration, SpO_2 monitoring**, and early mobilisation.
2. **Thromboprophylaxis**: **LMWH for 7 days after vaginal birth, or 6 weeks after caesarean section** (RCOG), plus mobilisation. Where LMWH is unaffordable, at least while in hospital and with early mobilisation, and discuss with seniors.
3. **Incentive spirometry** especially after CS; watch for ACS in the first 72 hours.
4. Continue **folic acid** and **penicillin**; check **Hb** before discharge.
5. **Restart hydroxyurea** after delivery only if **not breastfeeding** (or per haematology advice); breastfeeding is encouraged otherwise.
6. **Contraception** (WHO MEC): **progestogen-only methods (implant, DMPA, POP) and the LNG-IUD are category 1**. DMPA may even **reduce painful crises**. Combined hormonal contraception and copper IUD are category 2 (usable). Pregnancy spacing matters.
7. **Newborn**: if the partner is a carrier or unknown, arrange **newborn screening** (DBS for Hb electrophoresis/HPLC at the immunisation visit where available). SCD in the baby presents after about 6 months (dactylitis, anaemia, infection).
8. Link back to the **SCD clinic**.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Folic acid	**5 mg**	Oral	Once daily	Before conception, throughout pregnancy and after (lifelong in SCD)	Antagonises SP: see malaria box
Penicillin V (phenoxymethylpenicillin)	**250 mg** (some protocols 500 mg)	Oral	Twice daily	Throughout pregnancy	**Erythromycin 250 mg twice daily** if allergic
Aspirin (low-dose)	**75 mg**	Oral	Once daily at night	**From 12 weeks until 36 weeks**	Pre-eclampsia prophylaxis
Proguanil (malaria chemoprevention option)	**200 mg**	Oral	Once daily	Throughout pregnancy	Safe in pregnancy; give with folic acid; plus ITN

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
SP (IPTp) (alternative, unit protocol)	Standard IPTp dose	Oral	Monthly from 13 weeks	Through pregnancy	Withhold folic acid 5 mg for 1–2 weeks after each dose; **never with co-trimoxazole**
Iron	Only if **ferritin-proven deficiency**	Oral			Avoid routine FeFo
LMWH (e.g. enoxaparin)	**40 mg** (adjust for weight)	SC	Once daily	Every antenatal admission; postnatal **7 days** (vaginal) or **6 weeks** (CS)	Bleeding; timing with regional anaesthesia
Paracetamol	**1 g**	Oral or IV	6-hourly	During crisis	Max 4 g/day
Morphine (first-line opioid)	**2.5–5 mg IV** (or 5–10 mg SC / about 0.1 mg/kg), titrated; then e.g. oral morphine 10 mg 4-hourly	IV, SC, oral	Every 20–30 min until controlled, then 2–4-hourly	Until pain settles	Monitor RR, sedation and SpO_2; naloxone available; laxative and antiemetic with it; hypoventilation → ACS
Ibuprofen / diclofenac (NSAIDs)	Ibuprofen 400 mg; diclofenac 50 mg	Oral	8-hourly	Short course	**Only between 12 and 28 weeks** (avoid in the first trimester and after 28 weeks: premature closure of the ductus arteriosus, oligohydramnios); renal function
Codeine / tramadol	Codeine 30–60 mg; tramadol 50–100 mg	Oral	6-hourly	Step-down	For moderate pain or step-down
Pethidine	**Avoid**				Norpethidine accumulation causes **seizures**
IV fluids (0.9% saline or Ringer's lactate)	e.g. 1 L over 4–6 hours then reassess; oral intake about 3 L/day	IV / oral		While not drinking	Warm fluids; fluid balance chart; avoid overload
Oxygen	To SpO_2 94–98%	Nasal prongs or mask	Continuous	While SpO_2 below 95%	
Ceftriaxone	**2 g**	IV	Once daily	Fever / ACS course	Give after cultures; do not wait for mRDT if septic
Azithromycin (ACS)	**500 mg**	Oral	Once daily	ACS course	Or **erythromycin 500 mg 6-hourly**
Artesunate (severe malaria)	**2.4 mg/kg**	IV	0, 12, 24 h, then daily	Until able to take oral ACT	Uncomplicated malaria: oral artemether-lumefantrine (check UCG for the first trimester)
Red cells (top-up or exchange)	To restore, not exceed, baseline; ceiling about 10–11 g/dL	IV			HbS-negative, ABO/Rh (C, c, E, e)/Kell-matched, antibody-screened, CMV-negative
Hydroxyurea	**Stop** at least 3 months before conception				Restart postpartum only if not breastfeeding (or per haematology)

Referral criteria

Level	What should happen	Refer up when
HC III	Identify SCD at booking, give folic acid 5 mg, withhold routine iron, ITN, malaria prevention per protocol, **refer for hospital-based ANC**; treat fever promptly and refer crises	Every pregnant woman with SCD (booking) and every crisis → HC IV or hospital
HC IV	Initial crisis management: analgesia within 30 minutes (paracetamol, morphine or diclofenac 12–28 weeks), IV fluids, oxygen, mRDT and antimalarials, first dose IV ceftriaxone for fever	ACS, severe anaemia, uncontrolled pain, obstetric complication or no blood → General Hospital / RRH
General Hospital / RRH	Joint obstetric-haematology care, top-up transfusion, HDU, growth scans and CTG, planned delivery, epidural, postnatal LMWH	Need for exchange transfusion, ICU, ventilation, stroke, pulmonary hypertension, difficult cross-match → NRH
NRH (Mulago) and major centres	Exchange transfusion, ICU, national sickle cell clinic, echo, CTPA	n/a

Monitoring and follow-up

- **During a crisis**: pain score every 30 minutes until controlled; then 4-hourly pain, sedation, **RR, SpO_2**, temperature, BP, fetal heart; chest examination every 4 hours; **daily FBC** in severe crises; fluid balance chart.
- **Antenatal**: visits every 2–4 weeks (more often in the third trimester); BP, urinalysis, SFH, fetal heart, Hb as needed; urine culture each trimester; **growth scans every 4 weeks from 24 weeks**.
- **Postnatal**: crisis watch for 1–2 weeks; Hb before discharge; LMWH course; link back to the SCD clinic; contraception; newborn screening.

10. Complications

Disease-related

Maternal	Fetal / neonatal
Painful crises (most common; peak in third trimester and puerperium)	**Miscarriage**
Acute chest syndrome (leading cause of death)	**Fetal growth restriction** and low birth weight
Venous thromboembolism (DVT, PE)	**Preterm birth** (spontaneous and iatrogenic)
Pre-eclampsia/eclampsia	**Stillbirth** (several-fold increase)
Severe anaemia (sequestration, aplastic, hyperhaemolysis, malaria)	Fetal distress in labour; perinatal death
Infection: UTI/pyelonephritis, pneumonia, malaria, **puerperal sepsis**, pneumococcal sepsis	**Haemolytic disease of the fetus/newborn** (if maternal alloantibodies)
Stroke, pulmonary hypertension, heart failure	Fetal anaemia and hydrops (parvovirus B19)
Placental abruption, caesarean delivery, PPH	Inheritance of SCD (if partner carries HbS/HbC/β-thal)
Maternal death	

Treatment-related

Complication	Cause	Recognition and action
Alloimmunisation, delayed haemolytic reactions, **hyperhaemolysis**, transfusion-transmitted infection	Transfusion	Antibody screen, matched blood; stop transfusing in hyperhaemolysis; haematology advice
Hyperviscosity (crisis)	Over-transfusion	Keep Hb no higher than about 10–11 g/dL
Respiratory depression, hypoventilation → ACS	Opioids	RR, sedation, SpO_2; naloxone; incentive spirometry
Neonatal respiratory depression, withdrawal if prolonged use	Maternal opioids	Inform the neonatologist
Seizures	Pethidine (norpethidine)	Avoid pethidine
Ductus closure, oligohydramnios	NSAIDs after 28 weeks	Only 12–28 weeks
Pulmonary oedema	Over-hydration (pre-eclampsia, cardiac disease)	Fluid balance chart
Reduced malaria protection	Folic acid 5 mg with SP	Proguanil or withhold folate after SP

11. Prognosis

- Pregnancy in SCD carries a maternal mortality several times higher than in women without SCD, mainly from ACS, sepsis, severe anaemia and thromboembolism; African cohorts have poorer outcomes (maternal death, stillbirth, low birth weight) than high-income countries.
- Crisis risk peaks in the **third trimester and the puerperium**, and remains high for 1–2 weeks after delivery (about a quarter of women).
- With good care (planning, folic acid, malaria prevention, early analgesia, ACS vigilance, growth scans, planned delivery), **most women with SCD have a live baby**.
- **HbSC** women are often overlooked but share the risks of sequestration, ACS, VTE and retinopathy.

12. Prevention

- **Primary (genetic)**: know your genotype; partner testing (Hb electrophoresis) and genetic counselling before conception; prenatal diagnosis in few centres; newborn screening when the partner is a carrier or unknown.
- **Prevention of crises and complications in pregnancy**: avoid triggers (cold, dehydration, hypoxia, infection, acidosis); folic acid 5 mg; malaria chemoprevention and ITN; penicillin V and vaccines; aspirin from 12 weeks (pre-eclampsia); LMWH (VTE); incentive spirometry (ACS); early booking at a hospital.
- **Prevention of the next pregnancy's problems**: contraception (progestogen-only methods; DMPA may reduce crises), pregnancy spacing, planned pre-conception care.

13. Special Populations

- **Sickle cell trait (HbAS)**: normal pregnancy care with urine screening (more UTI) and partner counselling; **not** the SCD protocol.
- **HbSC and HbS/β-thalassaemia**: higher baseline Hb, spleen often still present → sequestration risk; retinal screening.
- **Pregnant on hydroxyurea**: stop it, reassure, detailed anomaly scan; do not advise termination on this basis alone.
- **HIV and co-trimoxazole**: **no SP**; co-trimoxazole provides malaria protection; continue folic acid 5 mg.

- **Previous ACS, stroke or worsening crises, or twin pregnancy**: consider a regular transfusion programme on specialist advice.
- **Pulmonary hypertension** (TR jet above 2.5 m/s): very high maternal risk; joint cardiac/obstetric care at a referral hospital.
- **Alloimmunised women**: early cross-match; risk of haemolytic disease of the fetus/newborn.
- **Breastfeeding**: encouraged; hydroxyurea restarted only if not breastfeeding (or per haematology advice).

14. Key Points

High-yield points

- Cold, dehydration, hypoxia, infection and acidosis make cells sickle: remove the trigger.
- Folic acid 5 mg; no routine iron.
- 5 mg folic acid antagonises SP: proguanil daily or withhold folate after SP.
- Stop hydroxyurea 3 months before conception.
- Analgesia within 30 minutes; morphine, never pethidine; NSAIDs only 12–28 weeks.
- Every crisis can become ACS: 4-hourly SpO_2 and RR, incentive spirometry.
- Compare every Hb with her baseline; reticulocytes separate the acute anaemias.
- Transfuse for indications, not routinely; HbS-negative, Rh/Kell-matched blood.
- Growth scans every 4 weeks from 24 weeks; deliver 38+0 to 40+0 weeks.
- Postnatal LMWH: 7 days (vaginal), 6 weeks (CS).

Numbers & doses

Item	Value
Uganda: trait / SCD in infants (Uganda Sickle Surveillance Study)	About **13%** / about **0.7%**
Folic acid	**5 mg** once daily, before conception and throughout (lifelong in SCD)
Penicillin V	**250 mg twice daily** (some protocols 500 mg); allergy: erythromycin 250 mg twice daily
Aspirin	**75 mg** at night from **12 to 36 weeks**
Hydroxyurea	Stop **at least 3 months** before conception
Malaria chemoprevention (unit-dependent)	**Proguanil 200 mg daily**, or monthly IPTp-SP from 13 weeks with folic acid withheld **1–2 weeks** after each dose; ITN always
Analgesia timing	Within **30 minutes** of arrival; reassess every **30 minutes**
Morphine	**2.5–5 mg IV** (or 5–10 mg SC), repeat every **20–30 min** until controlled
Paracetamol	**1 g** 6-hourly (max 4 g/day)
NSAIDs	Only **12–28 weeks**
Oxygen	If SpO_2 **below 95%** (RCOG 94%); ACS target **94–98%**
Oral fluids	About **3 L/day**
Incentive spirometry	**10 breaths every 2 hours** while awake
Fever / ACS antibiotics	**Ceftriaxone 2 g IV daily** (+ **azithromycin 500 mg daily** or erythromycin 500 mg 6-hourly for ACS)
Severe malaria	**IV artesunate 2.4 mg/kg** at 0, 12, 24 h, then daily

Item	Value
Transfusion triggers	Hb **below 6 g/dL** or fall of **2 g/dL or more** from baseline; ACS, stroke, multi-organ failure
Post-transfusion Hb ceiling	About **10–11 g/dL** (avoid hyperviscosity)
Blood matching	HbS-negative; **ABO, Rh (C, c, E, e), Kell** where possible
Pulmonary hypertension on echo	TR jet velocity **above 2.5 m/s**
Growth scans	Every **4 weeks from 24 weeks**
Delivery	**38+0 to 40+0 weeks**; SCD alone is not an indication for CS
LMWH	Every antenatal admission; postnatal **7 days** (vaginal) / **6 weeks** (CS) (RCOG); e.g. enoxaparin 40 mg SC daily
Genetics	SS × AA = **100% AS**; SS × AS = **50% SS**; AS × AS = **25% SS**

Exam traps

- **"Give routine FeFo like every other mother."** Wrong: SCD patients are usually iron-replete or overloaded. **Folic acid 5 mg**, iron only if ferritin is low.
- **"Give IPTp-SP with folic acid 5 mg."** Wrong: high-dose folate antagonises SP. Use **proguanil daily** or **withhold folic acid 1–2 weeks after SP**, per protocol.
- **"Pethidine for crisis pain."** Wrong: norpethidine causes **seizures**. Use **morphine**.
- **"Diclofenac at 34 weeks."** Wrong: NSAIDs only **12–28 weeks** (ductus closure, oligohydramnios).
- **"Hb 7 g/dL, she is always anaemic, no action."** Wrong until you compare with **her baseline**: a fall of 2 g/dL or more is an emergency.
- **"A positive sickling test confirms SCD."** Wrong: it is positive in **trait (AS)** too; confirm with **electrophoresis/HPLC**.
- **"Transfuse every pregnant woman with SCD prophylactically."** Wrong: **not routine**; transfuse for specific indications.
- **"Deliver her by CS because she has SCD" or "deliver to treat ACS."** Wrong: CS for **obstetric indications**; delivery does not treat ACS.

Questions seniors ask

On the ward round

1. **"What is her baseline Hb?"** Always know it; every decision about acute anaemia depends on the fall from baseline.
2. **"How long did it take her to get analgesia?"** It should be **under 30 minutes**; delays cause suffering and hypoventilation is a risk once large doses are finally given.
3. **"What was her SpO_2 and RR this morning, and what did the chest sound like?"** The daily ACS screen: 4-hourly SpO_2 and RR, auscultate the bases, incentive spirometry.
4. **"Her Hb has fallen from 7.5 to 5.2 and the reticulocytes are very low. What is it?" Aplastic crisis**, probably parvovirus B19: transfuse, send serology, scan the fetus for anaemia/hydrops.
5. **"What is your malaria prevention plan for her?"** Proguanil 200 mg daily with folic acid 5 mg (or SP with folic acid withheld after each dose per protocol) and an ITN; **no SP** if she is on co-trimoxazole.
6. **"When do you transfuse, and what blood do you ask for?"** Hb below 6 g/dL or fall of 2 g/dL or more, ACS, stroke, multi-organ failure; HbS-negative, ABO/Rh (C, c, E, e)/Kell-matched, antibody-screened; do not exceed about 10–11 g/dL.
7. **"How will she deliver?"** 38+0 to 40+0 weeks, vaginal unless obstetric indication; warm, hydrated, oxygenated (SpO_2 monitoring), continuous CTG, epidural encouraged, avoid prolonged labour, active third

stage.

8. **"What happens after delivery?"** Crisis risk stays high for 1–2 weeks; LMWH 7 days (vaginal) or 6 weeks (CS), incentive spirometry, folic acid and penicillin, progestogen-only contraception (DMPA may reduce crises).

Viva questions

1. **Why is folic acid 5 mg given, not 0.4 mg?** Chronic haemolysis increases red cell turnover and folate demand; deficiency leads to megaloblastic (aplastic-like) crises.
2. **Why no routine iron?** Most women with SCD are iron-replete or overloaded (recycled haem iron, transfusions). Give iron only if ferritin shows deficiency.
3. **What is acute chest syndrome and how do you treat it?** New CXR infiltrate plus fever/respiratory symptoms; oxygen, antibiotics (ceftriaxone + macrolide), analgesia, incentive spirometry, transfusion (top-up or exchange), HDU/ICU.
4. **Which analgesics are avoided?** Pethidine (seizures); NSAIDs before 12 and after 28 weeks.
5. **When do you transfuse?** Acute anaemia (Hb below 6 g/dL or a fall of 2 g/dL or more from baseline), ACS, stroke, multi-organ failure, pre-CS if low; not routinely.
6. **What do you stop before conception?** Hydroxyurea (3 months before), ACE inhibitors/ARBs, iron chelators.
7. **How do you monitor the fetus?** Growth scans every 4 weeks from 24 weeks; Dopplers if growth falls; CTG in labour.
8. **When and how should she deliver?** 38+0 to 40+0 weeks; vaginal unless obstetric indication; warm, hydrated, oxygenated, continuous CTG, regional analgesia.
9. **Postnatal thromboprophylaxis?** LMWH 7 days after vaginal birth, 6 weeks after CS.
10. **Best contraception?** Progestogen-only methods (DMPA may reduce crises), LNG-IUD, implant.
11. **Chance of an affected baby if the partner is AS?** 50% SS, 50% AS.

Mnemonics

Crisis triggers: "Cold, Dry, Low O_2, Bugs, Acid": cold, dehydration, hypoxia, infection (including malaria), acidosis (plus stress and exertion).

Antenatal package: "FAMPAS"

- **F**olic acid 5 mg
- **A**spirin 75 mg from 12 weeks
- **M**alaria prevention + ITN
- **P**enicillin V + vaccines
- **A**void iron, hydroxyurea, ACE inhibitors, pethidine
- **S**cans every 4 weeks from 24 weeks

Intrapartum: "4 W's": **W**arm, **W**et (hydrated), **W**ell-oxygenated, **W**atched (continuous CTG and SpO_2).

Painful crisis: "PHOTO": **P**ain relief within 30 minutes, **H**ydration, **O**xygen if SpO_2 below 95%, **T**riggers (infection, malaria) treated, **O**bstetric causes excluded (and **O**bserve for ACS).

Inheritance (Punnett squares)

Mother SS × partner AA	**A**	**A**
S	AS	AS
S	AS	AS

Result: 100% trait.

Mother SS × partner AS	A	S
S	AS	SS
S	AS	SS

Result: 50% trait, 50% SCD.

OSCE tips

- **Pre-conception counselling station**: cover risks, partner testing, folic acid 5 mg, stopping hydroxyurea, vaccines/penicillin, malaria prevention, avoiding triggers, early booking at a hospital, and the plan for aspirin and growth scans. Be honest but hopeful: most women with SCD who receive good care have a live baby.
- **Genetics**: draw the Punnett square: SS × AA = 100% AS; SS × AS = 50% SS, 50% AS; AS × AS = 25% SS, 50% AS, 25% AA.
- **Crisis station**: say "analgesia within 30 minutes", name morphine (not pethidine), the NSAID window (12–28 weeks), oxygen threshold, fluids, the trigger screen, and ACS surveillance; then the fetal heart.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What is the molecular defect in HbS?	A point mutation in the **beta-globin gene (codon 6)** replacing **glutamic acid with valine**. Deoxygenated HbS polymerises, making rigid sickled cells.
2	Name five triggers of sickling.	**Cold, dehydration, hypoxia, infection (including malaria) and acidosis** (also stress and exertion).
3	Why does pregnancy increase crises in SCD?	Dilutional anaemia and folate demand, **hypercoagulability**, higher oxygen demand, the **hypoxic placental bed**, more infections (UTI, malaria), vomiting/dehydration, and the stresses of labour and the puerperium.
4	Why do fetuses of mothers with SCD develop growth restriction?	Sickling and vaso-occlusion in the **uteroplacental circulation** cause placental infarction and insufficiency, reducing oxygen and nutrient delivery (plus maternal anaemia and hypoxia in crises).
5	Folic acid dose in SCD pregnancy and why?	**5 mg orally once daily**, before conception and throughout pregnancy: chronic **haemolysis** increases red cell turnover and folate demand.
6	Why avoid routine iron in SCD?	Iron from haemolysed cells is recycled and transfusions add iron, so women are usually **iron-replete or overloaded**. Give iron only if **ferritin** shows deficiency.
7	What should be stopped before conception in SCD?	**Hydroxyurea (at least 3 months before)**, **ACE inhibitors/ARBs**, and **iron chelators**.
8	Why is penicillin prophylaxis given in SCD, and the dose?	**Functional asplenia** (auto-infarction) predisposes to pneumococcal sepsis. **Penicillin V 250 mg orally twice daily** (erythromycin 250 mg twice daily if allergic), plus vaccines.
9	What is the SP–folic acid problem in Uganda?	WHO advises **not giving folic acid 5 mg or more with IPTp-SP** because high-dose folate antagonises SP. Options: **daily proguanil 200 mg** with folic acid, or SP with folic acid withheld for about 1–2 weeks after each dose, per unit protocol. Never omit malaria prevention.
10	Aspirin in SCD pregnancy?	**75 mg orally once daily from 12 weeks** (to about 36 weeks) to reduce **pre-eclampsia** risk.
11	Postnatal thromboprophylaxis in SCD (RCOG)?	**LMWH for 7 days after vaginal delivery** and **6 weeks after caesarean section**, plus LMWH during any antenatal admission.
12	First-line analgesia in a painful crisis?	Within **30 minutes**: **paracetamol 1 g** plus an **opioid (morphine)** titrated to effect; NSAIDs only **between 12 and 28 weeks**; **avoid pethidine** (norpethidine causes seizures).
13	Define acute chest syndrome.	A **new pulmonary infiltrate on chest X-ray** with fever and/or respiratory symptoms (cough, chest pain, tachypnoea, hypoxia). The leading cause of death in SCD.
14	Treatment of acute chest syndrome?	**Oxygen** (SpO_2 94–98%), **IV ceftriaxone + a macrolide** (azithromycin/erythromycin), analgesia, **incentive spirometry**, careful fluids, LMWH, **top-up or exchange transfusion**, HDU/ICU.
15	Indications for transfusion in SCD pregnancy?	**Acute anaemia**: Hb **below 6 g/dL** or a **fall of 2 g/dL or more from baseline**; **ACS**, **stroke**, multi-organ failure (exchange); pre-CS if Hb low; consider regular transfusion for previous severe complications or twins. **Not routinely prophylactic.**

#	Question	Answer
16	How do you tell an aplastic crisis from sequestration?	Both cause a sudden Hb fall. **Aplastic (parvovirus B19)**: **very low reticulocytes**, no organ enlargement. **Sequestration**: **high reticulocytes**, enlarging tender spleen/liver, shock.
17	Fetal surveillance schedule in SCD pregnancy?	Dating scan, anomaly scan at 18–22 weeks, then **growth scans every 4 weeks from 24 weeks** (with Doppler if growth falls), fetal movement awareness, continuous CTG in labour.
18	Timing and mode of delivery in SCD?	Offer delivery at **38+0 to 40+0 weeks** (IOL, or CS if indicated). **SCD alone is not an indication for CS.**
19	Intrapartum care principles in SCD?	**Warm, well-hydrated (fluid balance), oxygen if SpO_2 94% or below, continuous CTG**, early **regional analgesia**, avoid pethidine and GA, avoid prolonged labour, blood cross-matched, active third stage.
20	Best contraception for a woman with SCD (WHO MEC)?	**Progestogen-only methods** (implant, DMPA, POP) and the LNG-IUD are **category 1**; **DMPA may reduce painful crises**. COCs and copper IUD are category 2.
21	Partner is AS and the woman is SS. Risk to the baby?	**50% SS (disease), 50% AS (trait)**. If the partner is AA, all children are AS.
22	What bloods do you send in the first 5 minutes for a pregnant woman with SCD in crisis?	**FBC and reticulocytes** (compare with baseline Hb), **group and cross-match with antibody screen**, **mRDT/blood smear**, **blood cultures if febrile**, U&E and LFTs where available. Plus urine dipstick/culture, SpO_2, and a CXR if any chest sign. **Analgesia within 30 minutes** is given at the same time.
23	Why can a positive sickling (solubility) test not diagnose sickle cell disease?	It is positive in **both trait (HbAS) and disease (HbSS)**. The genotype needs **Hb electrophoresis or HPLC** (woman and partner). A woman with trait should not be put on the SCD protocol.
24	HbSC woman, baseline Hb 11 g/dL, now 7 g/dL with a tender enlarging spleen, tachycardia and high reticulocytes. Diagnosis and action?	**Splenic sequestration** (a fall of 4 g/dL). **Urgent transfusion** aiming to restore, not exceed, baseline (blood returns from the spleen as it resolves), senior and haematology input. HbSC is **not** mild in pregnancy.
25	Day 2 of a painful crisis at 32 weeks: T 38.5 °C, cough, RR 28, SpO_2 92%, basal crackles. What do you do?	Suspect **acute chest syndrome**: senior help, **oxygen to 94–98%**, **CXR** (with shielding) for a new infiltrate, cultures, **ceftriaxone 2 g IV daily + azithromycin 500 mg daily**, analgesia enough to breathe deeply, incentive spirometry, careful fluids, LMWH, cross-match for **top-up or exchange transfusion**, CTG. Delivery does not treat ACS.
26	How often should SpO_2 and RR be checked during an SCD crisis admission, and why?	At least **4-hourly** (continuous if falling), with daily chest auscultation, because **ACS often develops 1–3 days into a painful crisis**, especially with opioids and rib/back pain. A falling SpO_2 is the earliest sign.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 22-year-old with HbSS books at 10 weeks at a HC IV. The midwife prescribes the routine ANC pack: ferrous sulphate 200 mg + folic acid 0.4 mg daily and IPTp-SP from 13 weeks. What is the MOST appropriate modification?

A No change; the routine pack is adequate

B Folic acid 5 mg daily, stop routine iron unless ferritin shows deficiency, and plan malaria chemoprevention that accounts for the SP–folate interaction (e.g. daily proguanil) with an ITN

C Double the iron dose because she is anaemic

D Stop all malaria prevention because SP is contraindicated

E Give folic acid 0.4 mg twice daily and continue iron

2. A 25-year-old with HbSS on hydroxyurea 1 g daily discovers she is 7 weeks pregnant. She wants to continue the pregnancy. What is the BEST advice?

A Terminate the pregnancy because hydroxyurea is teratogenic

B Continue hydroxyurea because it prevents crises in pregnancy

C Switch hydroxyurea to iron chelation

D Halve the hydroxyurea dose

E Stop hydroxyurea now, reassure her, and arrange a detailed anomaly scan

3. A 24-year-old with HbSS at 32 weeks is admitted with severe back and thigh pain typical of her crises. Temperature 37.2 °C, SpO_2 97%, BP 110/70, uterus soft, fetal heart 140. What is the MOST appropriate immediate analgesic plan?

A IM pethidine 100 mg

B Oral ibuprofen 400 mg 8-hourly

C Paracetamol 1 g plus morphine titrated (e.g. 2.5–5 mg IV, repeated every 20–30 minutes) within 30 minutes, with antiemetic and laxative

D Wait for FBC results before giving analgesia

E Diclofenac 75 mg IM stat

4. A 21-year-old with HbSS, 2 days after admission for a painful crisis at 34 weeks, develops fever 38.4 °C, cough, pleuritic chest pain, RR 30 and SpO_2 89% on air. CXR shows new right lower lobe consolidation. What is the MOST appropriate management?

A Oxygen, IV ceftriaxone plus a macrolide, incentive spirometry, senior/HDU review, and urgent top-up or exchange transfusion

B Increase morphine and reassure

C Emergency caesarean section to improve ventilation

D Oral amoxicillin and discharge

E Furosemide 40 mg IV for pulmonary oedema

5. A 26-year-old with HbSC (baseline Hb 11 g/dL) at 29 weeks presents with left upper quadrant pain, pallor and dizziness. HR 128, BP 90/55. The spleen is palpable 6 cm below the costal margin. Hb 6.8 g/dL, reticulocytes high. What is the MOST likely diagnosis? Advanced

- A Aplastic crisis due to parvovirus B19
- B Iron deficiency anaemia
- C Placental abruption
- D Acute splenic sequestration
- E Vaso-occlusive crisis without anaemia

6. A 23-year-old with HbSS at 26 weeks has a Hb of 5.4 g/dL (baseline 7.8) after a febrile illness 1 week ago. She is tired but haemodynamically stable. The spleen and liver are not enlarged. Reticulocyte count is 0.2%. What is the MOST likely cause and a related concern? Advanced

- A Sequestration; risk of splenic rupture
- B Parvovirus B19 aplastic crisis; risk of fetal anaemia and hydrops
- C Hyperhaemolysis; risk of kernicterus
- D Folate excess; risk of neural tube defects
- E Iron overload; risk of cardiomyopathy

7. A 27-year-old with HbSS at 30 weeks presents with 'crisis pain' in the abdomen for 4 hours. She says it feels different. BP 150/95 (her usual is 105/65), the uterus is tense and tender, there is dark vaginal bleeding and the fetal heart is 100 bpm. What is the MOST likely diagnosis?

- A Vaso-occlusive crisis
- B Biliary colic
- C Acute pyelonephritis
- D Acute chest syndrome
- E Placental abruption

8. A 22-year-old with HbSS at 20 weeks has a painful crisis. She asks for ibuprofen, which helped before pregnancy. Which statement about NSAIDs in pregnancy is correct?

- A NSAIDs are contraindicated throughout pregnancy
- B NSAIDs are safest in the third trimester
- C NSAIDs may be used between 12 and 28 weeks, and should be avoided before 12 and after 28 weeks
- D NSAIDs are preferred over morphine at any gestation
- E NSAIDs can only be used IM

9. A 25-year-old with HbSS is at 38+2 weeks with normal fetal growth. She asks whether she needs a caesarean section because of her sickle cell disease. What is the BEST response?

- A No; plan induction of labour now (38–40 weeks) aiming for vaginal birth, with warmth, hydration, oxygen if needed, continuous CTG and early epidural; CS only for obstetric indications
- B Yes, SCD is an absolute indication for elective CS at 37 weeks
- C Await spontaneous labour until 42 weeks
- D CS under general anaesthesia is safest
- E Deliver only after a prophylactic exchange transfusion

10. A 28-year-old with HbSS delivered by caesarean section yesterday. What thromboprophylaxis plan is MOST appropriate?

A None; SCD protects against thrombosis
B LMWH for 7 days only
C Aspirin 75 mg for 6 weeks instead of LMWH
D LMWH (prophylactic dose) for 6 weeks with early mobilisation and hydration
E Warfarin for 3 months

11. A 24-year-old with HbSS at 28 weeks has Hb 7.2 g/dL (baseline 7.5), no symptoms of anaemia, and no complications. A colleague suggests a routine transfusion 'to prevent crises'. What is the BEST response?

A Agree; routine prophylactic transfusion is standard
B Disagree; routine prophylactic transfusion is not recommended; transfuse for specific indications such as acute anaemia (Hb below 6 or a fall of 2 g/dL or more), ACS or stroke
C Give IV iron instead
D Transfuse to a target Hb of 14 g/dL
E Transfuse only after delivery

12. A 23-year-old with HbSS in labour at 39 weeks, 6 cm dilated. SpO_2 is 92% on air, temperature 36.1 °C, she has vomited twice and is in severe pain. What is the MOST appropriate set of actions?

A Pethidine 100 mg IM and continue observation
B Artificial rupture of membranes and oxytocin augmentation only
C Immediate CS under general anaesthesia
D Restrict fluids to prevent pulmonary oedema
E Oxygen, warm environment and warm IV fluids with fluid balance chart, early epidural, continuous CTG, and review for infection/ACS

13. A 26-year-old with HbSS at 24 weeks attends ANC. Which fetal surveillance plan is MOST appropriate?

A No scans needed after the anomaly scan
B Weekly CTG from 24 weeks
C Growth scans every 4 weeks from 24 weeks, with umbilical artery Doppler if growth falls off
D Single scan at 36 weeks
E Amniocentesis to check fetal Hb

14. A 30-year-old with HbSS, 5 days after vaginal delivery, asks about contraception. She is breastfeeding and has had frequent crises. Which method is BEST for her?

A DMPA (injectable progestogen) or an implant
B Combined oral contraceptive pill
C No contraception needed while breastfeeding
D Hydroxyurea as contraception
E Bilateral tubal ligation is mandatory

15. A 20-year-old with HbSS at 16 weeks presents with fever 39 °C, rigors and joint pains. SpO_2 96%. She has not been using a mosquito net. What is the MOST appropriate immediate step?

A Treat as a painful crisis with analgesia alone
B Wait 24 hours to see if the fever settles
C Oral amoxicillin and review in 1 week
D mRDT/blood smear, blood cultures and urinalysis, then start IV ceftriaxone and antimalarials if positive (IV artesunate if severe), with analgesia, fluids and warmth
E Paracetamol only because fever is common in SCD

Answers & explanations

1. B · Folic acid 5 mg daily, stop routine iron unless ferritin shows deficiency, and plan malaria chemoprevention that accounts for the SP–folate interaction (e.g. daily proguanil) with an ITN

B is correct. SCD requires **folic acid 5 mg** (haemolysis), **no routine iron** (usually overloaded), and **malaria prevention** that avoids giving 5 mg folate with SP (e.g. proguanil) plus an ITN. **A** gives too little folate and unnecessary iron. **C**: SCD anaemia is haemolytic, not iron-deficient; iron worsens overload. **D**: malaria is a major crisis trigger and killer; prevention must continue. **E**: 0.8 mg folate is still inadequate and iron is inappropriate.

2. E · Stop hydroxyurea now, reassure her, and arrange a detailed anomaly scan

E is correct. Hydroxyurea should be stopped (ideally 3 months before conception); if conception occurs on it, **stop, reassure** (absolute risk appears low) and do a **detailed anomaly scan**. **A**: exposure alone is not an indication for termination. **B** and **D**: continued exposure during organogenesis is not recommended. **C**: chelators are also stopped in pregnancy and have a different purpose.

3. C · Paracetamol 1 g plus morphine titrated (e.g. 2.5–5 mg IV, repeated every 20–30 minutes) within 30 minutes, with antiemetic and laxative

C is correct. Analgesia within **30 minutes**; morphine is the opioid of choice with paracetamol; monitor sedation and RR. **A**: pethidine is avoided (norpethidine seizures). **B** and **E**: **NSAIDs are avoided after 28 weeks** (ductus arteriosus closure, oligohydramnios). **D**: never delay analgesia for results.

4. A · Oxygen, IV ceftriaxone plus a macrolide, incentive spirometry, senior/HDU review, and urgent top-up or exchange transfusion

A is correct. This is **acute chest syndrome**: oxygen, antibiotics covering typical and atypical organisms, analgesia with spirometry, careful fluids, and **transfusion** (exchange if severe hypoxia or worsening). **B**: more opioid may worsen hypoventilation. **C**: delivery does not treat ACS and an unstable hypoxic mother is at high risk from surgery. **D** is dangerously inadequate. **E**: there is no evidence of fluid overload; consolidation with fever indicates ACS.

5. D · Acute splenic sequestration

D is correct. Sudden Hb fall (11 to 6.8), **enlarging spleen**, shock and **high reticulocytes** point to **splenic sequestration**, which is more likely in **HbSC** (spleen not auto-infarcted). **A**: aplastic crisis has **low reticulocytes** and no organomegaly. **C**: abruption causes uterine pain/tenderness and bleeding, not splenomegaly. **B**: iron deficiency is gradual, with low reticulocytes. **E**: the dramatic anaemia and shock exceed a simple painful crisis. Management: urgent transfusion, resuscitation.

6. B · Parvovirus B19 aplastic crisis; risk of fetal anaemia and hydrops

B is correct. A fall in Hb with **very low reticulocytes** after a viral illness suggests **parvovirus B19 aplastic crisis**; parvovirus also infects fetal red cell precursors, causing **fetal anaemia and hydrops**, so arrange fetal ultrasound (MCA Doppler). She needs transfusion (Hb below 6). **A**: sequestration has high reticulocytes and organomegaly. **C**: hyperhaemolysis has high reticulocytes and jaundice. **D** and **E** do not explain an acute reticulocytopenic anaemia.

7. E · Placental abruption

E is correct. Tense, tender uterus, bleeding and fetal bradycardia indicate **abruption**, more likely here given relative hypertension (possible pre-eclampsia). The key lesson: **never assume abdominal pain in SCD pregnancy is a crisis**. **A** does not cause a woody uterus or bleeding. **C** causes fever and loin pain. **D** is a chest syndrome. **B** causes RUQ pain without uterine signs.

8. C · NSAIDs may be used between 12 and 28 weeks, and should be avoided before 12 and after 28 weeks

C is correct (RCOG). Before 12 weeks NSAIDs are associated with miscarriage; after 28 weeks they risk **premature ductus arteriosus closure** and **oligohydramnios**. At 20 weeks, ibuprofen is acceptable alongside paracetamol ± opioid. **A** is too restrictive. **B** is the opposite of the truth. **D**: severe crisis pain usually needs opioids. **E**: route is not the issue.

9. A · No; plan induction of labour now (38–40 weeks) aiming for vaginal birth, with warmth, hydration, oxygen if needed, continuous CTG and early epidural; CS only for obstetric indications

A is correct. RCOG advises delivery at **38+0 to 40+0 weeks**; SCD is **not** an indication for CS. **B** is wrong. **C**: prolonging pregnancy increases stillbirth and crisis risk. **D**: GA increases hypoxia/ACS risk; regional is preferred. **E**: prophylactic transfusion is not routine.

10. D · LMWH (prophylactic dose) for 6 weeks with early mobilisation and hydration

D is correct. RCOG recommends **LMWH for 6 weeks after caesarean section** (7 days after vaginal birth) in SCD. **A**: SCD is prothrombotic. **B** is the post-vaginal-delivery duration. **C**: aspirin is not adequate VTE prophylaxis. **E**: warfarin is not used for prophylaxis here.

11. B · Disagree; routine prophylactic transfusion is not recommended; transfuse for specific indications such as acute anaemia (Hb below 6 or a fall of 2 g/dL or more), ACS or stroke

B is correct. Her Hb is at baseline; **prophylactic transfusion is not routinely recommended** (risks: alloimmunisation, reactions, infection, iron overload, hyperhaemolysis). **A** is wrong. **C**: she is not iron deficient. **D**: Hb above about 10–11 causes hyperviscosity and crises. **E**: transfusion is given when indicated, at any time.

12. E · Oxygen, warm environment and warm IV fluids with fluid balance chart, early epidural, continuous CTG, and review for infection/ACS

E is correct. Intrapartum SCD care is **warm, hydrated, oxygenated and watched**: oxygen for SpO_2 below 94–95%, IV fluids given she is vomiting, **epidural** for pain, **continuous CTG**, and assess for ACS/infection given hypoxia. **A**: pethidine is avoided. **C**: GA worsens hypoxia risk and there is no obstetric indication. **D**: dehydration triggers sickling (use a fluid balance chart rather than restriction). **B** ignores her hypoxia, pain and dehydration.

13. C · Growth scans every 4 weeks from 24 weeks, with umbilical artery Doppler if growth falls off

C is correct. SCD causes placental insufficiency; RCOG recommends **serial growth scans every 4 weeks from 24 weeks**. **A** and **D** would miss evolving IUGR. **B**: CTG at 24 weeks is uninterpretable and not a growth test. **E**: amniocentesis tests genotype, not growth, and is not routine.

14. A · DMPA (injectable progestogen) or an implant

A is correct. Progestogen-only methods are **WHO MEC category 1** in SCD, compatible with breastfeeding, and **DMPA may reduce painful crises**. **B**: COCs are category 2 in SCD but oestrogen is avoided in the first 6 weeks postpartum while breastfeeding (and adds VTE risk). **C**: lactational amenorrhoea is unreliable unless strict criteria are met. **D**: hydroxyurea is not a contraceptive. **E**: sterilisation is her choice, not mandatory.

15. D · mRDT/blood smear, blood cultures and urinalysis, then start IV ceftriaxone and antimalarials if positive (IV artesunate if severe), with analgesia, fluids and warmth

D is correct. Fever in SCD is an emergency (**functional asplenia** plus malaria risk): test for malaria, culture blood and urine, and give **empirical IV antibiotics** promptly plus antimalarials when positive. **A**, **B**, **E** risk fulminant sepsis or severe malaria. **C** is inadequate empirical cover for suspected sepsis in an asplenic patient.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Pre-conception and booking in HbSS

Doreen, 24, with HbSS (baseline Hb 7.8 g/dL), attends the Mulago sickle cell clinic saying she wants a baby. She takes hydroxyurea 1 g daily, folic acid 5 mg and penicillin V. She had acute chest syndrome once at 17 years. Her husband has never been tested.

1. What history do you take, and why?

> Genotype and **baseline Hb**; crisis frequency, **previous ACS/stroke**, ICU admissions; **transfusion history and antibodies**; end-organ damage (breathlessness: PHT; kidney disease; eyes; leg ulcers; AVN); medications (**hydroxyurea**, ACEi, chelators); vaccination status; malaria prevention; previous pregnancies; partner status; social support and distance to a hospital.

2. What investigations do you arrange before conception?

> FBC, retics, ferritin, renal function and urine protein:creatinine, LFTs, red cell antibody screen/phenotype, **echo** (TR velocity for PHT), retinal exam, HIV/syphilis/HBsAg, and **partner Hb electrophoresis**.

3. What advice and changes do you make?

> **Stop hydroxyurea at least 3 months before trying to conceive**; continue **folic acid 5 mg** and **penicillin V 250 mg twice daily**; update pneumococcal, meningococcal, HiB, hepatitis B, influenza vaccines; plan malaria prevention and ITN; stop any ACEi or chelators. Counsel on risks (crises, ACS, VTE, pre-eclampsia, IUGR, stillbirth) and the plan to reduce them. If partner is **AS**: 50% chance of SCD baby; offer genetic counselling.

4. She books at 9 weeks. Outline her antenatal plan.

> Hospital-based ANC every 2–4 weeks (more often later). **Aspirin 75 mg from 12 weeks**. **No routine iron** (ferritin-guided). Malaria chemoprevention accounting for the **SP–folate interaction** (e.g. proguanil 200 mg daily) + ITN. Urine culture each trimester. Dating and anomaly scans, **growth scans every 4 weeks from 24 weeks**. LMWH during any admission. Education: hydrate (about 3 L/day), avoid cold, seek care early for fever, pain, cough or breathlessness.

5. How do you plan her delivery and postnatal care?

> Deliver at a hospital at **38+0 to 40+0 weeks** (IOL; CS only if indicated). Intrapartum: warm, hydrated, SpO_2 monitoring and oxygen if 94% or below, **continuous CTG**, **early epidural**, avoid pethidine and prolonged labour, blood cross-matched. Postnatal: LMWH (7 days after vaginal birth, 6 weeks after CS), spirometry, hydration, continue folic acid/penicillin, progestogen-only contraception (DMPA may reduce crises), **newborn screening** if partner is a carrier, restart hydroxyurea when not breastfeeding.

Take-away: Pre-conception planning (stop hydroxyurea, 5 mg folic acid, vaccines, partner testing) and a structured ANC package (aspirin, malaria prevention, no routine iron, serial growth scans) are what make SCD pregnancies safer.

Case 2: Painful crisis that becomes acute chest syndrome

Rose, 22, HbSS, G1 at 31 weeks, is brought to Jinja Regional Referral Hospital with 1 day of severe pain in her back and legs after a cold, rainy night. Temperature 37.8 °C, HR 108, RR 20, SpO_2 96%, BP 108/64. Uterus

soft, fetal heart 150.

1. What are your immediate priorities in the first 30 minutes?

> ABCDE; **analgesia within 30 minutes**: paracetamol 1 g + **morphine** titrated (e.g. 2.5–5 mg IV every 20–30 minutes until controlled), with antiemetic and laxative; **no NSAIDs** (31 weeks), **no pethidine**. Warm her. IV or oral fluids with fluid balance. Oxygen if SpO_2 below 95%. Bloods: FBC, retics, U&E, LFTs, cross-match/antibody screen, **mRDT**, blood culture; urinalysis/culture. Exclude obstetric causes (uterus soft, no bleeding, CTG).

2. What else goes on the admission plan?

> LMWH prophylaxis, **incentive spirometry** 10 breaths 2-hourly, continue folic acid, penicillin and aspirin, 4-hourly SpO_2, RR, temperature, pain and sedation scores, daily FBC, CTG monitoring, treat malaria if positive (IV artesunate if severe).

3. On day 2 she develops cough, chest pain, RR 32, SpO_2 88% on air, temperature 38.9 °C. Examination: bronchial breathing at the right base. What is happening and what are the differentials?

> **Acute chest syndrome** (typical timing, 1–3 days into a painful crisis, often after opioids). Differentials: bacterial pneumonia, **pulmonary embolism**, pulmonary oedema (fluid overload), atelectasis. Confirm with **CXR** (with shielding): new infiltrate.

4. How do you manage it?

> Senior help, HDU. **Oxygen** to SpO_2 94–98%. **IV ceftriaxone 2 g daily + azithromycin 500 mg daily** (or erythromycin). Analgesia enough to allow deep breathing, spirometry, careful fluids, LMWH. FBC: Hb 5.9 (baseline 7.2) so **transfuse** (top-up); **exchange transfusion** if she deteriorates or has multilobar disease (refer to a centre able to do it if needed). Continuous fetal monitoring; delivery is not a treatment for ACS.

5. She improves over 4 days. What follow-up is needed?

> Step down analgesia and antibiotics, repeat Hb, confirm growth on ultrasound (placental insult), increase ANC frequency, discuss a **transfusion programme** with haematology (recurrent severe complications), reinforce hydration, warmth and early presentation, plan delivery at a hospital with blood and HDU at 38 weeks or earlier if indicated.

> **Take-away:** Every painful crisis in pregnancy is a potential acute chest syndrome; monitor SpO_2 and RR closely and act fast with oxygen, antibiotics, spirometry and transfusion.

Case 3: Abdominal pain in SCD: crisis or something else?

Night shift at a HC IV: Esther, 27, HbSS, G3P2 at 34 weeks, arrives saying 'my sickle pain has started in my belly'. She is in severe pain and pale. HR 118, BP 138/88 (her usual is about 100/60), temperature 36.8 °C.

1. What must you think about before calling this a crisis?

> Obstetric: **placental abruption**, **labour**, **pre-eclampsia/HELLP** (BP high for her). Surgical/medical: pyelonephritis, appendicitis, cholecystitis (pigment stones), hepatic sequestration, bowel obstruction. SCD: vaso-occlusive (mesenteric) crisis, splenic/hepatic sequestration.

2. What examination findings and quick tests will guide you?

> Uterine tone and tenderness, vaginal bleeding, contractions, fetal heart; RUQ/epigastric tenderness, liver and spleen size, renal angle; urine dipstick (**protein**, nitrites, blood), Hb (HemoCue), mRDT, blood glucose. Findings: **tense tender uterus, dark bleeding, fetal heart 96 bpm, proteinuria 2+**.

3. What is the diagnosis and what do you do at the HC IV?

> **Placental abruption with likely pre-eclampsia** in a woman with SCD. Resuscitate: left lateral, oxygen, 2 large IV lines, warm fluids, bloods (Hb, grouping and cross-match, bedside clotting test), catheter. **$MgSO_4$** (loading dose) if severe features, antihypertensive if BP 160/110 or more. Analgesia (morphine). With a live fetus in distress, **urgent delivery by CS** at the HC IV if theatre and blood are available; otherwise stabilise and **refer urgently** with a midwife.

4. What SCD-specific precautions apply around the delivery?

> Blood cross-matched early (possible antibodies), **transfuse** given anaemia and bleeding risk; regional anaesthesia if clotting allows; keep warm, oxygenated and hydrated; avoid pethidine; active third stage and PPH preparedness; postnatal **LMWH for 6 weeks after CS** (once bleeding risk settles), spirometry, SpO_2 monitoring, watch for crisis/ACS; neonatal team for a preterm baby.

Take-away: In SCD pregnancy, abdominal pain is not a crisis until abruption, labour, pre-eclampsia and surgical causes have been excluded by examination and simple tests.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.